

🔑 🔑 Assignment: Sorting and Aggregating Data 🔑 🔑

READ THE DATASET

```
In [1]: import pandas as pd
```

```
In [2]: data = pd.read_csv('dataset/time_series.csv')
```

```
In [3]: data.head()
```

```
Out[3]:
```

	ID	Datetime	Count
0	0	25-08-2012 00:00	8
1	1	25-08-2012 01:00	2
2	2	25-08-2012 02:00	6
3	3	25-08-2012 03:00	2
4	4	25-08-2012 04:00	2

```
In [4]: data.dtypes
```

```
Out[4]: ID          int64
Datetime  object
Count     int64
dtype: object
```

Convert the Datetime column to pandas datetime object

```
In [5]: # WRITE YOUR CODE HERE
timedata = pd.to_datetime(data.Datetime, format='%d-%m-%Y %H:%M')
timedata.head()
```

```
Out[5]: 0    2012-08-25 00:00:00
1    2012-08-25 01:00:00
2    2012-08-25 02:00:00
3    2012-08-25 03:00:00
4    2012-08-25 04:00:00
Name: Datetime, dtype: datetime64[ns]
```

Create the following features using the Datetime feature

- day of month
 - month
 - day of year
 - hour
-

```
In [6]: # WRITE YOUR CODE HERE
day_of_month = timedata.dt.day
month = timedata.dt.month
day_of_year = timedata.dt.dayofyear
year = timedata.dt.year
```

Consider the following dataframe

```
In [7]: sample_df = pd.DataFrame({
    'id' : [1, 2, 3, 4, 5],
    'name' : [ 'Akash', 'Akansha', 'Naman', 'Pulkit', 'Anand'],
    'marks': [ 22, 23, 43, 41, 12]
})
sample_df
```

Out[7]:

	id	name	marks
0	1	Akash	22
1	2	Akansha	23
2	3	Naman	43
3	4	Pulkit	41
4	5	Anand	12

Sort the dataframe by marks column in descending order

```
In [8]: # WRITE YOUR CODE HERE
marks = sample_df.sort_values(by='marks', ascending=False)
marks
```

Out[8]:

	id	name	marks
2	3	Naman	43
3	4	Pulkit	41
1	2	Akansha	23
0	1	Akash	22
4	5	Anand	12

Consider another dataframe city

```
In [9]: city = pd.DataFrame({
        'name' : ['Akash', 'Akansha', 'Pulkit', 'Anand', 'Naman'],
        'city' : ['Delhi', 'Gurugram', 'Gurugram', 'Delhi', 'Noida']
    })
city
```

Out[9]:

	name	city
0	Akash	Delhi
1	Akansha	Gurugram
2	Pulkit	Gurugram
3	Anand	Delhi
4	Naman	Noida

Use the dataframe `sample_df` and `city` and find out the marks of students who live in Gurugram.

```
In [10]: ### Write your code here
city.merge(sample_df, how='left', on='name')
```

Out[10]:

	name	city	id	marks
0	Akash	Delhi	1	22
1	Akansha	Gurugram	2	23
2	Pulkit	Gurugram	4	41
3	Anand	Delhi	5	12
4	Naman	Noida	3	43

READ THE BIG MART SALES DATA

```
In [11]: big_mart_data = pd.read_csv('dataset/big_mart_sales.csv')
```

```
In [12]: big_mart_data.head()
```

Out[12]:

	Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Identifier
0	FDA15	9.30	Low Fat	0.016047	Dairy	249.8092	C
1	DRC01	5.92	Regular	0.019278	Soft Drinks	48.2692	C
2	FDN15	17.50	Low Fat	0.016760	Meat	141.6180	C
3	FDX07	19.20	Regular	0.000000	Fruits and Vegetables	182.0950	C
4	NCD19	8.93	Low Fat	0.000000	Household	53.8614	C

Find out the number of unique Item_Type in each of the Outlet_Identifier

```
In [13]: # WRITE YOUR CODE HERE
big_mart_data.Item_Type.unique()
```

```
Out[13]: array(['Dairy', 'Soft Drinks', 'Meat', 'Fruits and Vegetables',
                'Household', 'Baking Goods', 'Snack Foods', 'Frozen Foods',
                'Breakfast', 'Health and Hygiene', 'Hard Drinks', 'Canned',
                'Breads', 'Starchy Foods', 'Others', 'Seafood'], dtype=object)
```

Find out the average Item_Outlet_Sales for each category of Outlet_Size

```
In [14]: # WRITE YOUR CODE HERE
big_mart_data.groupby('Outlet_Size')['Item_Outlet_Sales'].mean()
```

```
Out[14]: Outlet_Size
High      2298.995256
Medium    2681.603542
Small     1912.149161
Name: Item_Outlet_Sales, dtype: float64
```

Select only the categorical variables from the dataframe and use `get_dummies` function to convert those into the numerical ones.

```
In [15]: # WRITE YOUR CODE HERE
big_mart_cat = big_mart_data.select_dtypes(include='object')
big_mart_cat
```

```
Out[15]:
```

	Item_Identifier	Item_Fat_Content	Item_Type	Outlet_Identifier	Outlet_Size	Outlet_Location_
0	FDA15	Low Fat	Dairy	OUT049	Medium	1
1	DRC01	Regular	Soft Drinks	OUT018	Medium	1
2	FDN15	Low Fat	Meat	OUT049	Medium	1
3	FDX07	Regular	Fruits and Vegetables	OUT010	NaN	1
4	NCD19	Low Fat	Household	OUT013	High	1
...	...	...	...	...	...	...
8518	FDF22	Low Fat	Snack Foods	OUT013	High	1
8519	FDS36	Regular	Baking Goods	OUT045	NaN	1
8520	NCJ29	Low Fat	Health and Hygiene	OUT035	Small	1
8521	FDN46	Regular	Snack Foods	OUT018	Medium	1
8522	DRG01	Low Fat	Soft Drinks	OUT046	Small	1

8523 rows × 7 columns

```
In [16]: pd.get_dummies(big_mart_cat)
```

Out[16]:

	Item_Identifier_DRA12	Item_Identifier_DRA24	Item_Identifier_DRA59	Item_Identifier_DRB01
0	0	0	0	0
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0
...	...	...	...	...
8518	0	0	0	0
8519	0	0	0	0
8520	0	0	0	0
8521	0	0	0	0
8522	0	0	0	0

8523 rows × 1600 columns

